SDSU | San Diego State University

Department of Engineering

ME341: Engineering Design: Mechanical Components

Fall Semester 2025

Student Design Project: Torsion Bar

Section 01

Authors: Thomas Bonar

Instructor: Prof. Ricardo Ayala

November 17, 2025

Table of Contents

Introduction 3
Requirements 4
Torsion Bar Shaft Design 4
Shaft Spline Design and Analysis 5
Bearing Support Lug Design 6
Torsion Bar Fatigue Analysis 7
Fixed Coupling Analysis 8
Axle Design and Analysis 9
Lever Arm Sizing and Analysis 10
Final Questions 12
Conclusion 12
Appendix 13

Table of Figures

Figure 1: Torsion Bar Schematic 3
Figure 2: Torsion Bar Detail 4
Figure 3: Torsion Bar 5
Figure 4: Bearing Support Lug Detail 5
Figure 5: Fatigue life (S-N) Curve 6
Figure 6: S-N Curve 7
Figure 7: Fixed Coupling Detail 7
Figure 8: Shear Force Diagram 8
Figure 9: Bending Moment Diagram 9
Figure 10: Axle Detail 9
Figure 11: Lever Arm Detail 10
Figure 12: Range of Lever Arm Motion 11

Table of Tables

Table 1: Torsion Bar Analysis Results 3
Table 2: Spline Details 6
Table 3: Spline Stress Analysis Results 6
Table 4: Lug Details 8
Table 5: List of Variables and Equations 14

Introduction:

The following report covers the design and analysis of a conceptual torsion bar design. A suspension system for a wheel and tire assembly requires a steel torsion bar to be subjected to a specific loading. A solid steel axle is mounted on the lower section of a lever arm which fits on the steel torsion bar through a series of splines. The steel torsion bar is bolted to the side wall on one end and held up by a mounted bearing support near the other end as illustrated in Figure 1. A bearing sleeve ensures a snug fit around the main shaft as well as rotational independence. The axle and lever arm are suspended under tension which subjects the assembly to torsional, shear, and compressive forces. The following design aims to satisfy functionality demands while meeting the necessary safety requirements. Economical and efficiency factors were also considered in the design process. All dimensions are conducted in US Customary System (USCS) units.

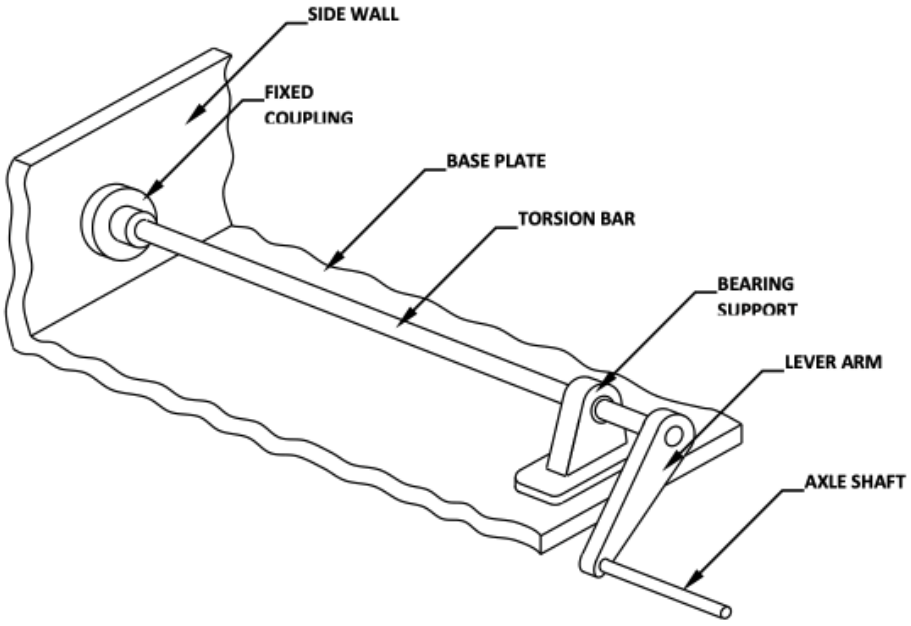


Figure 1: Torsion Bar Schematic

Requirements:

Because the torsion bar assembly is fixed in place, added forces on the axle put stress on the materials of the assembly. Thus, the torsion bar requires a specific relationship between the type of steel and the inner and outer diameters to maintain a static design factor of safety of 1.15 in torsion. The yield strength of the material must also hold for the splines which connect the main shaft to the wall as well as the lever arm. Design factors of safety of 1.05 in the splines must hold for shear, compression, and shaft forces. Additional strain applied through the bearing support requires a fatigue analysis through DE-Gerber criteria such that it passes another design factor of safety of 1.10. This fatigue analysis can be visualized in a $S_a \; vs. \; S_m$ fatigue curve and a fatigue life (S-N) curve laid out in figures 5 and 6, respectively.

To hold up the torsion bar shaft, a bearing support must be added near the end fixed to the base plate. The bearing is required to be an asymmetrical aluminum casting. Stress analysis on the dimensions of bore diameter, outside radius, and lug thickness must ensure a design factor of safety of 3.75 throughout the part. A chosen Symmco sleeve bearing must be trimmed and fitted within the support lug to ensure a snug connection to the shaft while maintaining rotational independence. The bearing support must be bolted to the aluminum base plate through two steel mounting bolts. Analysis of the chosen material properties and joint stiffness must pass a 1.15 factor of safety for yield, overload, and joint separation in the bearing support as well as in the bolts.

The only external force applied to the system occurs through the solid steel axle as depicted in Figure 10. Thus, a suitable steel material and a diameter must be chosen to fulfill the design factor of 1.25 without overengineering the part and driving up costs. Analysis into the shear and bending-moment stresses on the axle must be illustrated in diagrams to assess vulnerabilities. As the axle experiences a downward force, the entire torsion body will yield and bend under the strain. Thus, vertical suspension travel and torque minimums and maximums must be evaluated to determine the load capabilities of the chosen materials and dimensions.

Because the torsion bar is fixed to the wall, applied torsion is directed into the wall dispersed through steel bolts. Thus, the reactions at the wall must be found to adequately determine the number and quality of bolts such that stability and safety may be secured.

Torsion Bar Shaft Design:

For the torsion bar shaft, 4340 steel quenched and tempered at 600℉ was chosen for the main shaft due to its relatively high yield strength. Despite the extra cost for higher-grade steel, less material will be required which makes it similar in price to a thicker shaft made of cheaper steel With an outer diameter of 1.625 in and an inner diameter of 0.75 in, the shaft falls within the confines of a 3 in outer diameter and a necessary hollow aspect. These dimensions on the specified steel enable the shaft to endure torque forces of almost 6000 $in \cdot lbf$ without yield. Table 1 lists the characteristics and capabilities of the torsion bar.

Table 1: Torsion Bar Analysis Results

Max Applied Torque	Min Applied Torque	Max Shear Stress	Static Factor of Safety	Min Shear Stress
5,962 $in \cdot lbf$	1,491 $in \cdot lbf$	7,413 psi	15.5 > 1.15	1853 psi

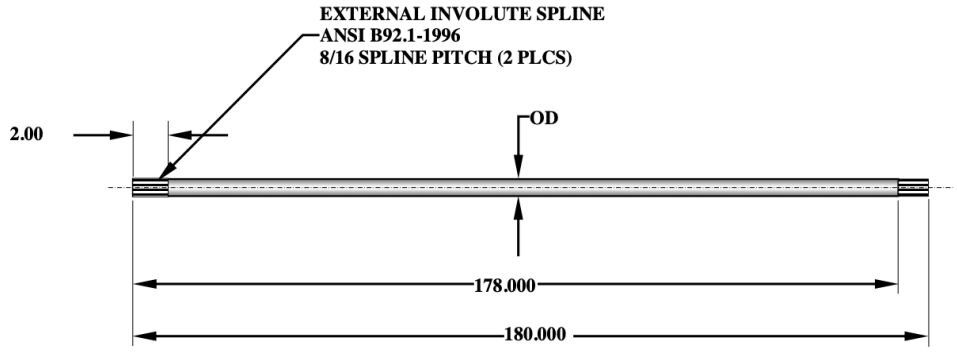


Figure 2: Torsion Bar Detail

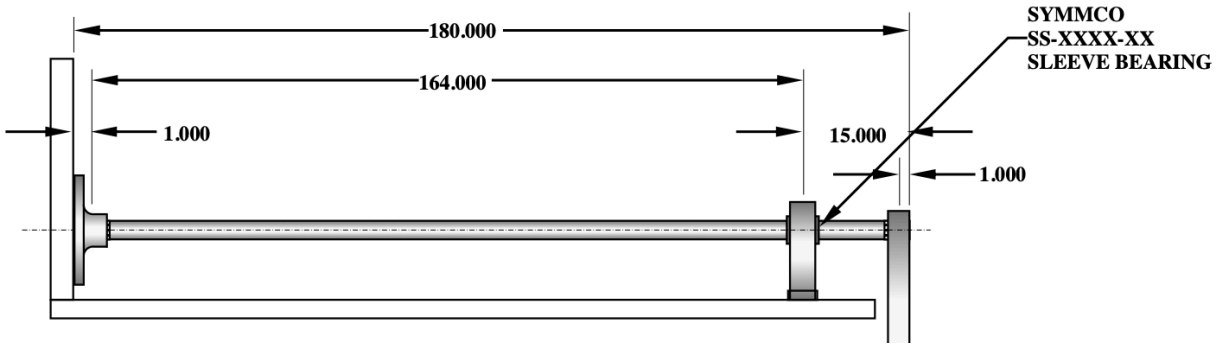


Figure 3: Torsion Bar

Shaft Spline Design and Analysis:

With an outer diameter of 1.625 in on the shaft, the end splines have coordinated spline dimensions listed in the ANSI B92-1 8-16 SPLINE PITCH Table which are also listed in tables 2 and 5. The dimensions all comfortably surpass their respective design factors of safety while still considering costs and efficiency by minimizing material.

Table 2: Spline Details

External Spline Root Diameter	Bore Diameter	Number of Teeth	Pitch Diameter	Face Width	Radial Tooth Height
1.317 in	0.75 in	12	1.5 in	2 in	0.4375 in

Table 3: Spline Stress Analysis Results

Shaft Shear Stress	FOS	Shear Stress in Teeth	FOS	Compressive Stress	FOS
14,855 psi	1.38 > 1.05	10,599 psi	1.93 > 1.05	757.1 psi	273.4 > 1.05

Bearing Support Lug Design:

The bearing support lug was chosen to have an outer radius of 1.5 in, an inner diameter of 2.0 in, and a lug thickness of 0.1 in as represented by X.XX, YYY, and Z.ZZ in Figure 4, respectively. The material chosen for the bearing support lug is a T5 333.0 cast aluminum alloy because of its strong and

light-weight properties. The size dimensions chosen ensure the bearing comfortably satisfies the factor of safety of 3.75 while also minimizing the material required. Although the lug is relatively thin, the characteristics make it capable of bearing over 24,000 psi of stress before damaging. Further details on the lug are listed in tables 4 and 5.

The sleeve bearing chosen was the Symmco SS-5264-16 with a thickness of 1in. This sleeve has an outer diameter of 2.004 in which will fit snug within the 2 in diameter of the bearing lug bore hole. The thickness of the lug is only 0.1 in which will require 90% the 1 in lug to be cut off before assembly. The 1 in sleeve is the smallest, cheapest product in the SS-5264-16 category.

Table 4: Lug Details

Lug Radius	Lug Thickness	Bearing Stress	Lug Strength	Factor of Safety
1.5 in	0.1 in	24,043 psi	34000 psi	9.9589 > 3.75

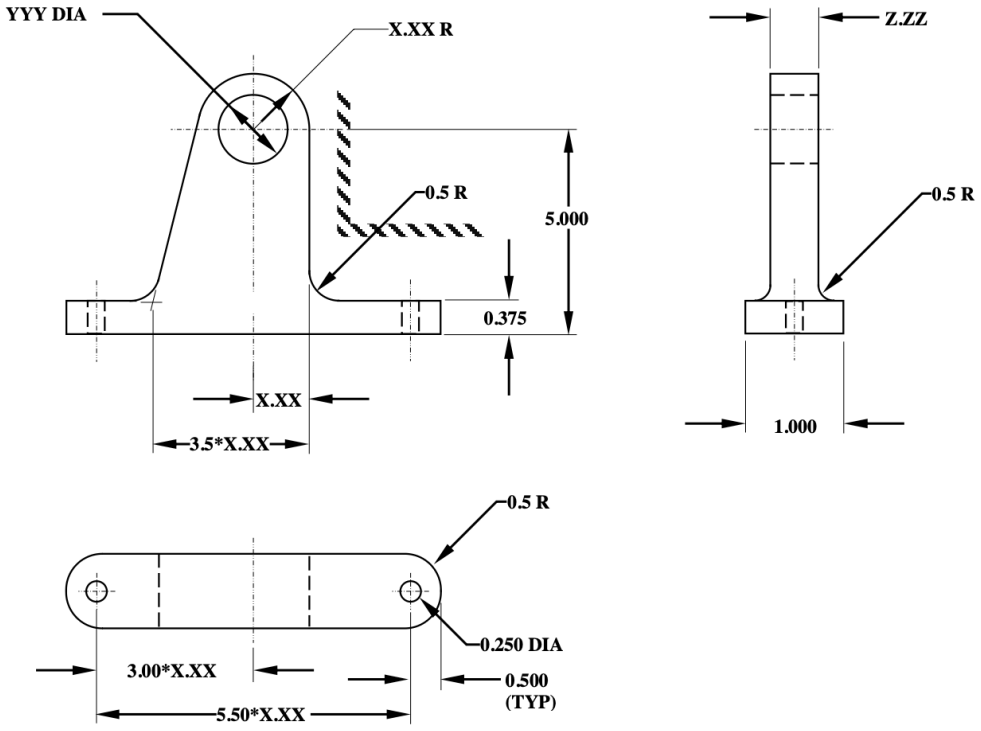


Figure 4: Bearing Support Lug Detail

Torsion Bar Fatigue Analysis:

The torsion bar has a Gerber bearing support fatigue factor of safety of 4.7 which well exceeds the 1.10 minimum requirement. The analysis for the fatigue life cure and Sa vs. Sm fatigue curve are represented in figures 5 and 6, respectively.

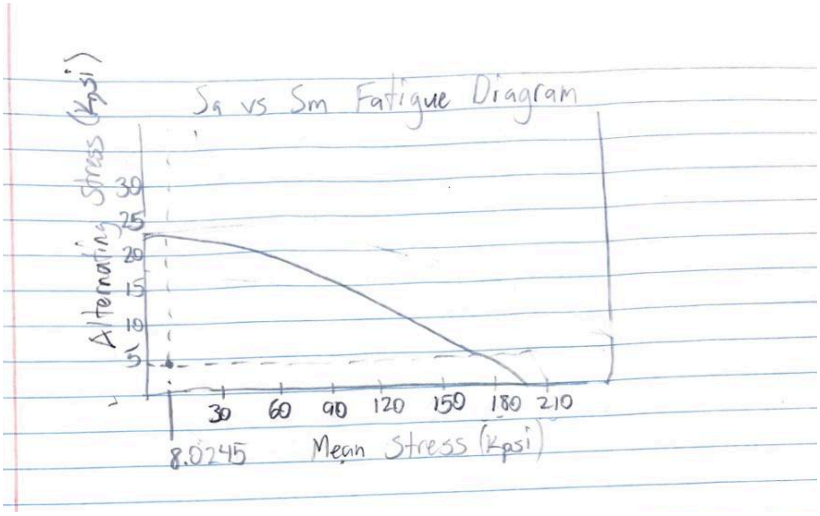


Figure 5: Fatigue life (S-N) Curve

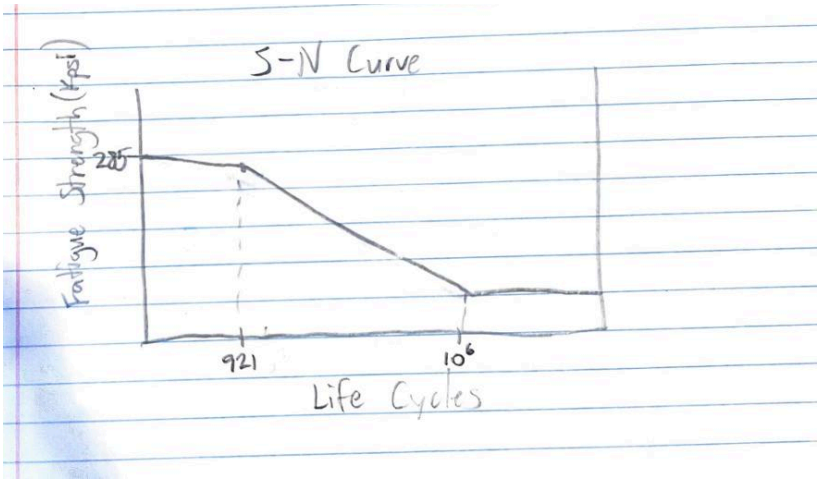


Figure 6: S-N Curve

Fixed Coupling and Bolt Analysis:

The main shaft is joined with the side wall through a coupling with splines and steel bolts. Reactions at the side wall were found to be 721 lbf in both the x and y directions at maximum load capacity. To ensure stability and safety, four ¼ in. SAE Grade 2 steel bolts were chosen for the mount with a #11 course series bolt thread. These bolts comfortably exceed the 1.75 design factor of safety as listed in Table 5 without demanding high-cost, high-quality material or finish. In fact, these bolts may be used for both the fixed coupling and the bearing support lug which both fit into 0.25 in bore holes making it easier to source, store, and replace parts.

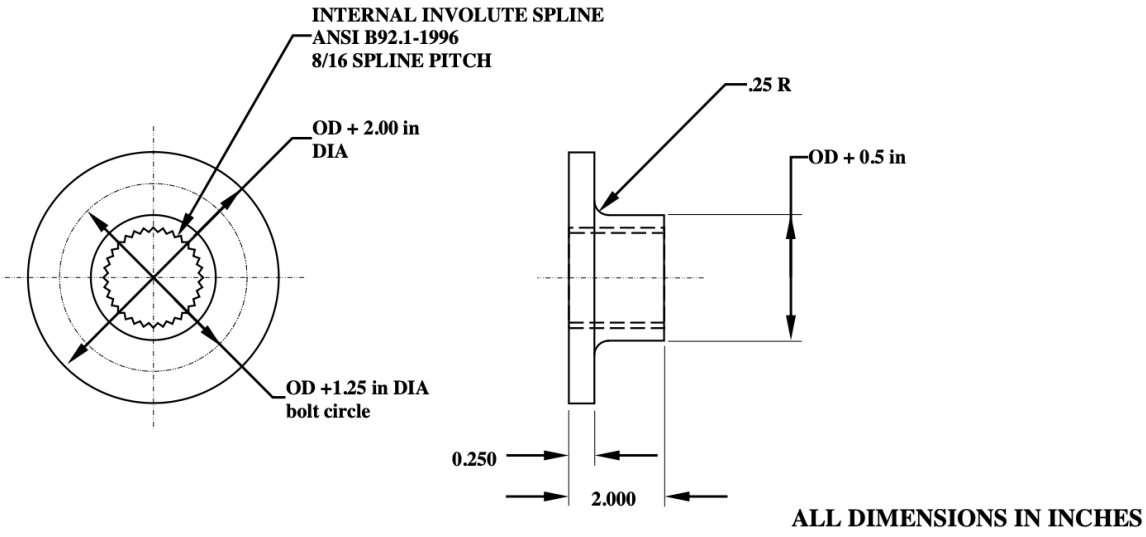


Figure 7: Fixed Coupling Detail

Axle Design and Analysis:

The axle will utilize SAE 1006 cold-drawn steel and will be manufactured to a solid diameter of 1.25 in. A factor of safety of 1.25 requires a minimum diameter of 1.092 in, but this was rounded to 1.25 in for easier manufacturing. The vertical suspension travel was found to be 1.924 in from the static positioning. With this material and dimensions, the load intensity can incur up to 311 lbf before deformation or $w_0(x)$ of 19.441 lbf*in across the length of the axle in its specified load sequence.

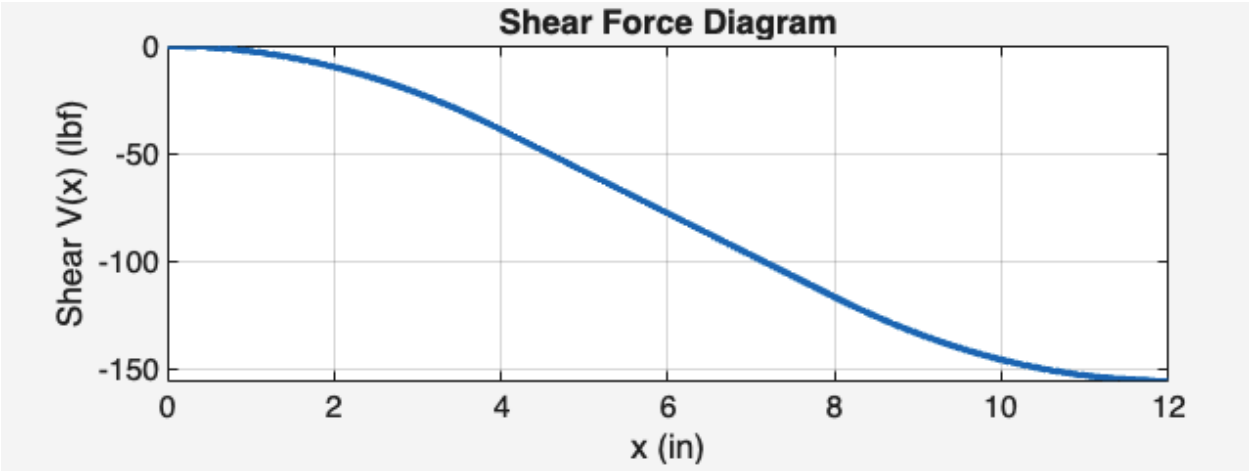


Figure 8: Shear Force Diagram

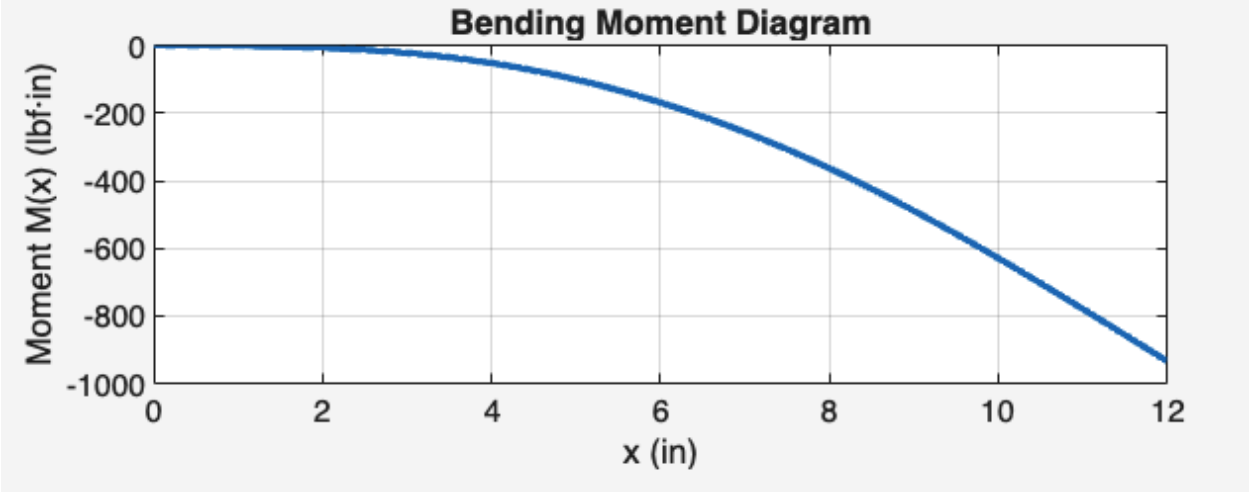


Figure 9: Bending Moment Diagram

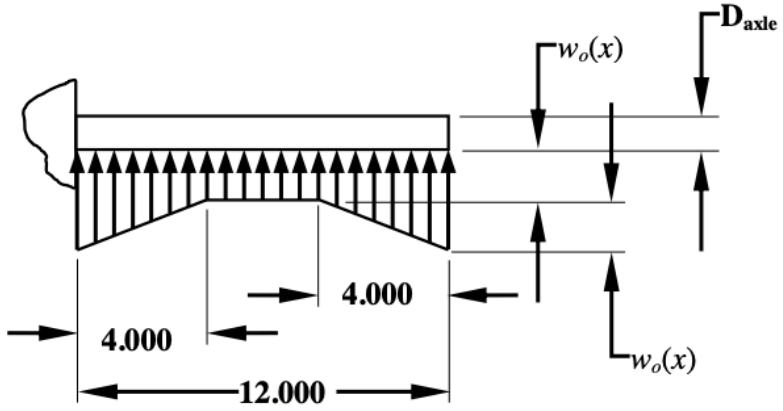


Figure 10: Axle Detail

Lever Arm Sizing and Analysis:

The lever arm is almost fully defined from the given parameters, but the splines and diameters of the two holes match the axle diameter dimensions and the shaft diameter dimensions.

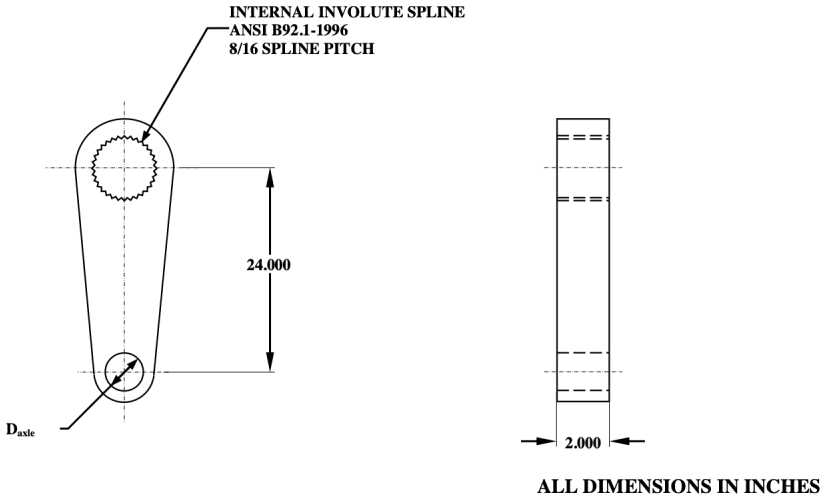


Figure 11: Lever Arm Detail

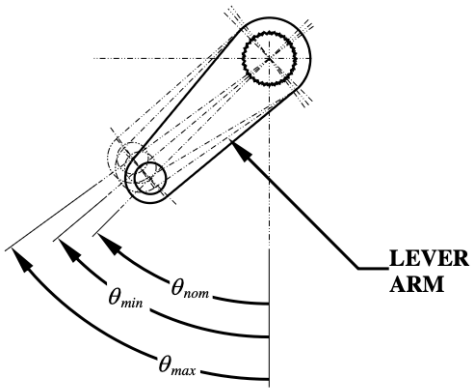


Figure 12: Range of Lever Arm Motion

Final Questions:

1. How you made informed judgements that considered the impact of your design solution on the environment and society (both locally and globally)?

In almost all aspects of the design, materials and dimensions were carefully chosen to minimize the materials necessary while ensuring the safety factors were met and slightly exceeded. By reducing the materials necessary, steel plants and other manufacturers would not need to produce as much which would ultimately cut down on global mining and energy consumptions. Even on a small scale, this could reduce carbon emissions and habitat destruction around the world.

2. How economic factors were considered in your design solution?

In the same way, material volume was minimized such that less material would need to be purchased. In addition, bolts with course finishes were chosen as seen on pg. 7 under the Fixed Coupling Section which are easier to manufacture and thus reducing the overall cost of the build.

3. How ethical and professional considerations were factored into your design solution?

The straightforward design prioritizes effectiveness and efficiency. Simple, cheap bolts and common steels chosen in the Torsion Bar Shaft Design Section on page 4 make it easy and cheap to source materials while ensuring a long-lasting product

Conclusion:

The overall design for the torsion bar assembly fulfills the requirements outlined. All factors of safety are met and exceeded to ensure quality and safety of product use. The assembly presents an efficient design as it competently takes the specified load without being overengineered and driving up prices.

Appendix:

Table 5: List of Variables and Equations

Symbol	Description	Origin	Value
Given Values			
$\theta_{Nom,\,d}$	Nominal Angle	Given	45°
$\theta_{Min,\,d}$	Minimum Angle of Deflection	Given	47°
$\theta_{Max,\,d}$	Maximum Angle of Deflection	Given	53°
$\Delta\theta_{Min,\,d}$	Minimum Deflection	Given	2°
$\Delta\theta_{Max,\,d}$	Maximum Deflection	Given	8°
$L_{Torsion\ Bar,\,E}$	Effective Torsion Bar Length	Given	176 in
$L_{Torsion\ Bar}$	Total Torsion Bar Length	Given	180 in
G	Modulus of Rigidity for Steel	Given	$11.5x10^{6}\ psi$
$n_{Spline,\,Min}$	Minimum Factor of Safety for Splines	Given	1.05
F_e	Effective Face Width	Given	2 in
R	Reliability	Given	99.9%
L_{LA}	Lever Arm Length	Given	24 in
L_{BW}	Bearing to Wall Length	Given	164 in
k_{cf}	Lug Casting Factor	Given	0.5
D_{Bolt}	Bolt Diameter	Given	0.25 in
t_{BS}	Thickness of Bearing Support Base	Given	0.375 in

t_{BP}	Thickness of Aluminum Base Plate	Given	0.4375 in
E_{Steel}	Modulus of Elasticity for Steel	Given	$30x10^6\ psi$
$E_{Aluminum}$	Modulus of Elasticity for Aluminum	Given	$10.3x10^6\ psi$
α	Half Apex Angle	Given	30°
L_{Axle}	Axle Length	Given	12 in
$n_{Axle,\,Min}$	Minimum Axle Factor of Safety	Given	1.25
Shaft			
D_{out}	Outer Diameter	Chosen	1.625 in
D_{in}	Inner Diameter	Chosen	0.75 in
$S_{y,\,Sh}$	Yield Strength of 4340 Q&T Steel @ 600°F	Table A-20	230 kpsi
$\theta_{Min,\,r}$	Minimum Angle of Deflection	$\theta_{Min,\,d}\left(\frac{\pi}{180}\right) = (47)\left(\frac{\pi}{180}\right)$	0.8203 rad
$\theta_{Max,\,r}$	Maximum Angle of Deflection	$\theta_{Max,\,d}\left(\frac{\pi}{180}\right) = (53)\left(\frac{\pi}{180}\right)$	0.925 rad
$\Delta\theta_{Min,\,r}$	Minimum Deflection	$\Delta\theta_{Min,\,d}\left(\frac{\pi}{180}\right) = (2)\left(\frac{\pi}{180}\right)$	0.0349 rad
$\Delta\theta_{Max,\,r}$	Maximum Deflection	$\Delta\theta_{Max,\,d}\left(\frac{\pi}{180}\right) = (8)\left(\frac{\pi}{180}\right)$	0.1369 rad
J	Moment of Inertia	$\frac{\pi}{32}\left(D_{out}^{\ 2} - D_{in}^{\ 2}\right)$	0.6535
T_{Min}	Minimum Torque	$\frac{\Delta\theta_{Min,\,r}(G)(J)}{L_{Torsion\ Bar,\,E}} = \frac{(0.0349)(11.5x10^6)(0.6535)}{(176)}$	1,490.5 $in \cdot lbf$
T_{Max}	Maximum Torque	$\frac{\theta_{Max,\,d}(G)(J)}{L_{Torsion\ Bar,\,E}} = \frac{(0.6535)(11.5x10^6)(0.6535)}{(176)}$	5,962.1 $in \cdot lbf$
τ_{Min}	Minimum Shear Stress	$\frac{T_{Min}\left(\frac{D_{out}}{2}\right)}{J} = \frac{1490.5\left(\frac{1.625}{2}\right)}{0.6535}$	1,853.2 psi

τ_{Max}	Maximum Shear Stress	$\frac{T_{Max}(\frac{D_{out}}{2})}{J} = \frac{1490.5(\frac{1.625}{2})}{0.6535}$	7,412.7 psi
$n_{d,\,max}$	Shaft Static Factor of Safety in Torsion	$\frac{\frac{S_{y,Sh}}{2}}{\tau_{Max}}$	15.5139 > 1.15
Splines			
D_{RE}	External Spline Root Diameter	ANSI B-92-1 Spline Chart	1.317 in
D_b	Bore Diameter	ANSI B-92-1 Spline Chart	0.75 in
N	Number of Splines	ANSI B-92-1 Spline Chart	12
D_P	Pitch Diameter	ANSI B-92-1 Spline Chart	1.5 in
h	Radial Tooth Height	$\frac{D_{out}-D_{in}}{2} = \frac{1.625-0.75}{2}$	0.4375 in
$S_{S,\,Sh}$	Shear Stress in Hollow Shaft	$\frac{16(T_{Max})(D_{RE})}{\pi(D_{RE}^{\,4}-D_b^{\,4})} = \frac{16(5962.1)(1.317)}{\pi(1.317^4-0.75^4)}$	14,855 psi
K_a	Spline Application Factor for Light Shock Input/Intermittent Shock Load	From Spline Slides, Slide 31	1.8
L_f	Fatigue Life Factor after 10^6 cycles	From Spline Slides, Slide 33	0.4
$S_{s\,Sh,\,Min}^{\,a}$	Allowable Shear Stress of Hollow Shaft	$\frac{n_{Spline,\,Min}(S_{s,Sh})(K_a)}{L_f} = \frac{(1.05)\,(14855)(1.8)}{(0.4)}$	70,190 psi
$S_{s\,Sh}^{\,a}$	Minimum Allowable Shear Stress	$0.4(S_{y,\,Sh})(1000) = 0.4(230)(1000)$	92,000 psi
$S_{c\,Sh,\,Min}^{\,a}$	Minimum Allowable Compressive Stress	$0.45(S_{y,\,Sh})(1000) = 0.45(230)(1000)$	103,500 psi
t_e	Chordal Thickness at Pitch Line	$\frac{D_p}{2(N)} = \frac{1.5}{2(12)}$	0.0625 in
K_m	Load Distribution Factor		1.0

$S_{s,\,teeth}$	Spline Teeth Shear Stress	$\frac{4(T_{Max})(K_m)}{(D_p)(N)(F_e)(t_e)} = \frac{4(5962.1)(1.0)}{(1.5)(12)(2)(0.0625)}$	10,599 psi
$S_{c,\,teeth}$	Spline Teeth Compressive Stress	$\frac{4(T_{Max})(K_m)}{(D_p)(N)(F_e)(h)} = \frac{4(5962.1)(1.0)}{(1.5)(12)(2)(0.4375)}$	757.0898 psi
$n_{S,\,Teeth}$	Spline Teeth Shear Factor of Safety	$\frac{(S_{s_{Sh}}^{a})(L_f)}{(S_{s,\,teeth})(K_a)} = \frac{(92000)(0.4)}{(10599)(1.8)}$	1.9289 > 1.05
n_{Spline}	Spline Factor of Safety	$\frac{(S_{s_{Sh}}^{a})(L_f)}{(S_{s,\,Sh})(K_a)} = \frac{(92000)(0.4)}{(14855)(1.8)}$	1.3763 > 1.05
$n_{S,\,Compression}$	Spline Teeth Compressive Stress Factor of Safety	$\frac{(S_{c_{Sh,\,Min}}^{a})(9)(L_f)}{(S_{c,\,teeth})(K_a)} = \frac{(103500)(9)(0.4)}{(757.09)(1.8)}$	273.415 > 1.05
S-N Curve Fatigue Analysis for Shaft			
a_f	Surface Condition Constant (Machined)	From Ch6, Slide 29 Chart	2.70
$S_{Ult,\,S}$	Ultimate/Tensile Stress for 4340 Q&T Steel @ 600℉	From Table A-21	250 kpsi
b_f	Surface Factor Exponent (Machined)	From Ch6, Slide 29	-0.265
k_a	Surface Condition Modification Factor	$a_f(S_{Ult,\,S})^{b_f} = 2.7(250)^{-0.265}$	0.625
k_b	Size Modification Factor	$(\frac{D_{out}}{0.3})^{-0.107}$	0.8346
S'_e	Rotating Beam Specimen Endurance	From Ch6, Slide 25 Chart	100 kpsi
k_c	Load Modification Factor for Torsion	From Ch6, slide 32	0.59
k_d	Temperature Modification Factor	Not Applicable	1.0
k_e	Reliability Modification Factor	From Ch6 slide 33	0.753
k_f	Miscellaneous-Effects	Not Applicable	1.0

	Modification Factor		
S_e	Endurance Limit at Critical Location	$(S'_e)(k_a)(k_b)(k_c)(k_d)(k_e)(k_f) = (100)(0.625)(0.8346)(0.59)(1.0)(0.753)(1.0)$	23.1765
σ_m	Mean Stress	$\left\|\frac{\tau_{Max}+\tau_{Min}}{2}\right\|(0.001) = \left\|\frac{7412.7+1853.2}{2}\right\|(0.001)$	4.6329 kpsi
σ_a	Alternating Stress	$\left\|\frac{\tau_{Max}-\tau_{Min}}{2}\right\|(0.001) = \left\|\frac{7412.7-1853.2}{2}\right\|(0.001)$	2.7789 kpsi
f_f	Fatigue Strength Fraction	$1.06 - 0.0028(S_{Ult,S}) + (6.8x10^{-6})(S_{Ult,S})^2 = 1.06 - 0.0028(250) + (6.8x10^{-6})(250)^2$	0.7913
a	Amplitude	$\frac{(f_f)(S_{Ult,S})^2}{S_e} = \frac{(0.7913)(250)^2}{23.1765}$	1,688.3
b	Exponent to Cycles	$(-\frac{1}{3})ln\frac{(f_f)(S_{Ult,S})}{S_E} = (-\frac{1}{3})ln(\frac{(0.7912)(250)}{23.1765}$	-0.7147
σ_1	Principal Stress 1 (Pure Shear)	τ_{Max}	7.4127 kpsi
σ_2	Principal Stress 2 (Pure Shear)	$-\tau_{Max}$	-7.4127 kpsi
σ_3	Principal Stress 3 (Pure Shear)		0 kpsi
σ'_{Max}	Von Mises Stress	$(\frac{(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2}{2})^{\frac{1}{2}} = (\frac{(7.4127--7.4127)^2+(7.4127-0)^2+(0-7.4127)^2}{2})^{\frac{1}{2}}$	12.8392 kpsi
σ'_m	Von Mises Mean Stress	$\sqrt{3}(\sigma_m) = \sqrt{3}(4.6329)$	8.0245 kpsi
σ'_a	Von Mises Alternating Stress	$\sqrt{3}(\sigma_a) = \sqrt{3}(2.7789)$	4.8147 kpsi
N_f	Number of Cycles to Failure	$(\frac{\sigma'_{Max}}{a})^{(1/b)} = (\frac{12.8392}{1688.3})^{(1/-0.7147)}$	921.8497
S_f	Basquin Fatigue Strength Equation	$a(N_f)^b = (1688.3)(921.8497)^{-0.7147}$	285.8511 kpsi
n_G	Gerber Factor of Safety	$\frac{-(\frac{\sigma'_a}{S_e})+\sqrt{(\frac{\sigma'_a}{S_e})^2+4(\frac{\sigma'_m}{S_{Ult,S}})^2}}{2(\frac{\sigma'_m}{S_{Ult,S}})^2} =$	4.7040 > 1.10

		$\frac{-(\frac{4.8147}{23.1765})+\sqrt{(\frac{4.8147}{23.1765})^2+4(\frac{8.0245}{250})^2}}{2(\frac{8.0245}{250})^2}$	
S'_m	Gerber Design Equation for Alternating vs. Mean Curve	$(S_{Ult,S})\sqrt{1-(\frac{\sigma'_a}{S'_e})} = (250)\sqrt{1-(\frac{4.8147}{100})}$	243.9074
Bearing Support Lug			
$S_{Ult,A}$	Ultimate Stress of Lug (Cast Aluminum 333.0 T5)	From Table A-24	34000 psi
r_L	Lug Radius	Chosen	1.5 in
D_L	Lug Hole Diameter	Chosen	2.0 in
t_L	Lug Thickness	Chosen	0.10 in
F_{LA}	Force on Lever Arm	$\frac{T_{Max}}{(L_{LA})sin(\theta_{max,r})} = \frac{5962.1}{(24)sin(0.925)}$	311.0557 lbf
F_B	Force on Bearing	$\frac{-(F_{LA})(L_{Torsion\ Bar})}{L_{BW}} = \frac{-(311.0557)(180)}{164}$	-341.4026 lbf
P	Load on Lug	$\lvert F_B \rvert$	341.4026 lbf
w_L	Width of Lug	$3.5(r_L) = 3.5(1.5)$	5.25 in
L_L	Length of Lug	$\frac{w_L}{2} - r_L = \frac{5.25}{2} - 1.5$	1.125 in
L_{DBH}	Distance Between Bolt Holes	$5.5(r_L) = 5.5(1.5)$	8.25 in
L_{CBH}	Center of Bolt Holes	$0.5(L_{DBH}) = 0.5(8.25)$	4.125 in
θ_L	Load Angle	$(\frac{\pi}{180})tan^{-1}(\frac{L_{CBH}}{L_{DBH}}) = (\frac{\pi}{180})tan^{-1}(\frac{4.125}{8.25})$	26.5651°
$\frac{P_\theta}{P_0}$	Offset Loading Correction	From Ch3, Slide 9 Chart	0.9447
D_{Bo}	Bearing Outer Diameter	From Symmco Oilite PDF	2.004 in
D_{Bi}	Bearing Inner Diameter	From Symmco Oilite PDF	1.629 in

L_B	Bearing Length	From Symmco Oilite PDF	1.0 in
$\frac{R}{D}$	Ratio of Lug Radius to Bore Diameter	$\frac{r_L}{D_{Bo}} = \frac{1.5}{2.004}$	0.7485
$\frac{W}{D}$	Ratio of Lug Width to Bore Diameter	$\frac{w_L}{D_{Bo}} = \frac{5.25}{2.004}$	2.6198
$\frac{S_{br}}{S_{Ult,A}}\|_{\frac{R}{D}}$	R/D Design Curve	$\frac{8(\frac{R}{D})-4}{3+2(\frac{R}{D})} = \frac{8(0.7485)-4}{3+2(0.7485)}$	0.4421
$\frac{S_{br}}{S_{Ult,A}}\|_{\frac{W}{D}}$	W/D Design Curve	$-1.04 + 1.0637(\frac{W}{D}) - 0.0250(\frac{W}{D})^2$ $= -1.04 + 1.0637(2.6198) - 0.0250(2.6198)^2$	1.5751
$S_{br}\|_{\frac{R}{D}}$	Allowable Bearing Stress for R/D	$(S_{Ult,A})(\frac{P_\theta}{P_0})(\frac{R}{D}) = (34000)(0.9447)(0.7485)$	24,043 psi
$S_{br}\|_{\frac{W}{D}}$	Allowable Bearing Stress for W/D	$(S_{Ult,A})(\frac{P_\theta}{P_0})(\frac{W}{D}) = (34000)(0.9447)$ (2.6198)	84,149 psi
$S'_{br}\|_{\frac{R}{D}}$	Allowable Bearing Stress Due to Casting Factor for R/D	$(S_{br}\|_{\frac{R}{D}})(k_{cf}) = (24043)(0.5)$	12,021 psi
$S'_{br}\|_{\frac{W}{D}}$	Allowable Bearing Stress Due to Casting Factor for W/D	$(S_{br}\|_{\frac{W}{D}})(k_{cf}) = (84149)(0.5)$	42,075 psi
σ_{Lug}	Lug Bearing Stress	$\frac{P}{(t_L)(D_L)} = \frac{341.4026}{(0.1)(2)}$	1,707 psi
n_{Lug}	Lug Factor of Safety	$\frac{(S_{Ult,A})(k_{cf})}{\sigma_{Lug}} = \frac{(34000)(0.5)}{1707}$	9.9589 > 3.75
$n_{\frac{R}{D}}$	R/D Factor of Safety	$\frac{S'_{br}\|_{\frac{R}{D}}}{\sigma_{Lug}} = \frac{12,021}{1707}$	7.0423
$n_{\frac{W}{D}}$	W/D Factor of Safety	$\frac{S'_{br}\|_{\frac{W}{D}}}{\sigma_{Lug}} = \frac{42075}{1707}$	24.6482
Bearing Support Mounting Bolts			
t_W	Washer Thickness	Chosen	0.032 in
h_{nut}	Nut Height	Chosen	0.21875 in

A_m	Minor Diameter Area	From Bolt Dimensions Chart	0.0269 in^2
A_t	Tensile Stress Area	From Bolt Dimensions Chart	0.0318 in^2
$S_{P,2}$	Bolt Proof Strength	From SAE Bolt Strength Chart	55,000 psi
L_T	Threaded Length	$2(D_{Bolt}) + \frac{1}{4} = 2(0.25) + \frac{1}{4}$	0.75 in
A_{MD}	Major Area Diameter of Bolt	$\frac{\pi(D_{Bolt})^2}{4} = \frac{\pi(0.25)^2}{4}$	0.0491 in^2
L_G	Grip Length	$2(t_w) + t_{BS} + t_{BP} = 2(0.032) + (0.375) + 0.4375$	0.8765 in
L_{Fa}	Fastener Length (Rounded to Standard)	$L_G + h_{nut} = 0.8765 + 0.21875$	1.095→1.25 in
A_U	Area of Unthreaded Portion	A_{MD}	0.0491 in^2
L_{UG}	Length of Unthreaded Portion in Grip	$L_{Fa} - L_T = 1.25 - 0.75$	0.3453 in
$k_{b,Lug}$	Fastener Stiffness	$\frac{(A_{MD})(A_t)(E_{Steel})}{(A_{MD})(L_G)+(A_{MD})(L_{UG})} = \frac{(0.0491)(0.0318)(30x10^6)}{(0.0491)(0.8765)+(0.0491)(0.3453}$	780,850
k_t	Spring Rate of Threaded Portion	$\frac{(A_m)(E_{Steel})}{L_G} = \frac{(0.0269)(30x10^6)}{0.8765}$	920,710
k_d	Spring Rate of Unthreaded Portion	$\frac{(A_m)(E_{Steel})}{L_{UG}} = \frac{(0.0269)(30x10^6)}{0.3453}$	4,265,400
α_{Rad}	Half Apex Angle, Radians	$\alpha(\frac{\pi}{180}) = 30(\frac{\pi}{180})$	0.5236 rad
t_1	Frustum Thickness 1	t_W	0.032 in
t_2	Frustum Thickness 2	t_{BS}	0.375 in
t_3	Frustum Thickness 3	$\frac{L_G}{2} - t_1 - t_2 = \frac{0.8765}{2} - 0.032 - 0.375$	0.0312 in
t_4	Frustum Thickness 4	$t_{BP} - t_3 = 0.4375 - 0.0312$	0.4062 in
t_5	Frustum Thickness 5	t_W	0.032 in

D_{F1}	Frustum Diameter 1	$1.5(D_{Bolt}) = 1.5(0.25)$	0.375 in
D_{F2}	Frustum Diameter 2	$D_{F1} + 2(t_1 + t_2)tan(\alpha_{Rad}) = 0.375 + 2(0.032 + 0.375)tan(0.5236)$	0.845 in
D_{F3}	Frustum Diameter 3	$D_{F1} + 2(t_1 + t_2 + t_3)tan(\alpha_{Rad}) = 0.375 + 2(0.032 + 0.375 + 0.0312)tan(0.5236)$	0.881 in
D_{F4}	Frustum Diameter 4	$D_{F1} + 2(t_5 + t_4)tan(\alpha_{Rad}) = 0.375 + 2(0.032 + 0.4062)tan(0.5236)$	0.881 in
D_{F5}	Frustum Diameter 5	$1.5(D_{Bolt}) = 1.5(0.25)$	0.375 in
K_1	Member Stiffness 1	$\frac{0.5774(\pi)(E_{Steel})(D_{Bolt})}{ln(\frac{(1.155(t_W)+D_{F1}-D_{Bolt})(D_{F1}+D_{Bolt})}{(1.155(t_W)+D_{F1}+D_{Bolt})(D_{F1}-D_{Bolt})})} = \frac{0.5774(\pi)(30x10^6)(0.25)}{ln(\frac{(1.155(0.032)+0.375-0.25)(0.375+0.25)}{(1.155(0.032)+0.375+0.25)(0.375-0.25)})}$	67,489,000
K_2	Member Stiffness 2	$\frac{0.5774(\pi)(E_{Aluminum})(D_{Bolt})}{ln(\frac{(1.155(t_2)+D_{F2}-D_{Bolt})(D_{F2}+D_{Bolt})}{(1.155(t_2)+D_{F2}+D_{Bolt})(D_{F2}-D_{Bolt})})} = \frac{0.5774(\pi)(10.3x10^6)(0.25)}{ln(\frac{(1.155(0.375)+0.845-0.25)(0.845+0.25)}{(1.155(0.375)+0.845+0.25)(0.845-0.25)})}$	21,862,000
K_3	Member Stiffness 3	$\frac{0.5774(\pi)(E_{Aluminum})(D_{Bolt})}{ln(\frac{(1.155(t_3)+D_{F3}-D_{Bolt})(D_{F3}+D_{Bolt})}{(1.155(t_3)+D_{F3}+D_{Bolt})(D_{F3}-D_{Bolt})})} = \frac{0.5774(\pi)(10.3x10^6)(0.25)}{ln(\frac{(1.155(0.0312)+0.881-0.25)(0.881+0.25)}{(1.155(0.0312)+0.881+0.25)(0.881-0.25)})}$	192,950,000
K_4	Member Stiffness 4	$\frac{0.5774(\pi)(E_{Aluminum})(D_{Bolt})}{ln(\frac{(1.155(t_4)+D_{F4}-D_{Bolt})(D_{F4}+D_{Bolt})}{(1.155(t_4)+D_{F4}+D_{Bolt})(D_{F4}-D_{Bolt})})} = \frac{0.5774(\pi)(10.3x10^6)(0.25)}{ln(\frac{(1.155(0.4062)+0.881-0.25)(0.881+0.25)}{(1.155(0.4062)+0.881+0.25)(0.881-0.25)})}$	22,360,000
K_5	Member Stiffness 5	K_1	67,489,000
K_m	Total Member Stiffness	$(\frac{1}{K_1} + \frac{1}{K_2} + \frac{1}{K_3} + \frac{1}{K_4} + \frac{1}{K_5}) = (\frac{1}{67,489,000} + \frac{1}{21,862,000} + \frac{1}{192,950,000} + \frac{1}{22,360,000} + \frac{1}{67,489,000})$	7,981,900
C	Stiffness Constant	$\frac{k_{b,Lug}}{k_{b,Lug}+k_m} = \frac{780,850}{780,850+7,981,900}$	0.0891
P_b	Portion of Load Taken	$(C)(P) = (0.0891)(341.4026)$	30.4222 psi

	by Bolt		
P_m	Portion of Load Taken by Members	$P - P_b = 341.4026 - 30.4222$	310.9804 psi
F_p	Proof Load	$(A_t)(S_{P,2}) = (0.0318)(55000)$	1,749 lbf
F_i	Preload	$(0.75)(F_p) = 0.75(1749)$	1,311.8 lbf
F_b	Resultant Bolt Load	$F_i + P_b = 1311.8 + 30.4222$	1,342.2 lbf
F_m	Resultant Load on Connected Members	$P_b - F_i = 30.4222 - 1311.8$	-1,281.3 < 0
σ_{BS}	Tensile Stress Under Service Load	$\frac{F_b}{A_t} = \frac{1342.2}{0.0318}$	42.2067 kpsi
P_0	Value of External Load that Would Cause Joint Separation	$\frac{F_i}{1-C} = \frac{1311.8}{1-0.0891}$	1,440.1 psi
n_0	Factor of Safety Against Joint Separation	$\frac{P_0}{P} = \frac{1440.1}{341.4026}$	4.2181 > 1.15
n_L	Load Factor of Safety	$\frac{(S_{P,2})(A_t)-F_i}{P_b} = \frac{(55000)(0.0318)-(1311.8)}{30.4222}$	14.3727 > 1.15
n_p	Yielding Factor of Safety	$\frac{S_{P,2}}{\sigma_{BS}} = \frac{55000}{42.2067}$	1.3031 > 1.15
Axle Shaft Equations			
d	Axle Diameter	Chosen from $d_{Axle,\,Min}$	1.25 in
$W_{Max}(x)$	Maximum Axle Shaft Load	$\frac{T_{Max}}{(L_{LA})sin(\theta_{Max,r})} = \frac{5962.1}{(24)sin(0.925)}$	311.0557 lbf
$W_{Min}(x)$	Minimum Axle Shaft Load	$\frac{T_{Max}}{(L_{LA})sin(\theta_{Min,r})} = \frac{5962.1}{(24)sin(0.8203)}$	84.9179 lbf
$\theta_{Static,r}$	Static Axle Position	$\theta_{Nom,d}(\frac{\pi}{180}) = 45(\frac{\pi}{180})$	0.7854 rad
ΔL	Change in Length Between Loaded and Unloaded Conditions	$(L_{Axle})cos(\theta_{Static,r}) - (L_{Axle})cos(\theta_{Max,r}) = (24)cos(0.7854) - (24)cos(0.925)$	1.2635 in

$w_{0,X}$	Maximum Total Load (From Trapezoidal Distributed Load Geometry	$\frac{W_{Max}(x)}{16} = \frac{311.0557}{16}$	19.441 $lbf \cdot in$
W	Total Load on Axle	$W_{Max}(x)$	311.0557 lbf
$\bar{x}$	Center of Axle	$\frac{L_{Axle}}{2} = \frac{12}{2}$	6 in
M_{Max}	Maximum Bending Moment	$W(\bar{x}) = 311.0557(6)$	1,866.3 $lbf \cdot in$
M_{Min}	Minimum Bending Moment	$W_{Min}(x)(6) = (84.9179)(6)$	509.5074 $lbf \cdot in$
M_a	Alternating Moment	$\frac{M_{Max} - M_{Min}}{2} = \frac{1866.3 - 509.5074}{2}$	678.4133 $lbf \cdot in$
M_m	Mean Moment	$\frac{M_{Max} + M_{Min}}{2} = \frac{1866.3 + 509.5074}{2}$	1,187.9 $lbf \cdot in$
T_A	Alternating Torque	$\frac{T_{Max} - T_{Min}}{2} = \frac{5962.1 - 1490.5}{2}$	2,235.8 $lbf \cdot in$
T_m	Mean Torque	$\frac{T_{Max} + T_{Min}}{2} = \frac{5962.1 + 1490.5}{2}$	3,726.3 $lbf \cdot in$
K_{fn}	Fatigue Notch Factor (Bending and Axial)		1.0
K_{fs}	Fatigue Notch Factor (Torsion)		1.0
A	DE Goodman Criteria, Alternating	$\sqrt{4(K_{fn}M_a)^2 + 3(K_{fs}T_a)^2} = \sqrt{4(1x678.4133)^2 + 3(1x2235.8)^2}$	4,103.3 $lbf \cdot in$
B	DE Goodman Criteria, Mean	$\sqrt{4(K_{fn}M_m)^2 + 3(K_{fs}T_m)^2} = \sqrt{4(1x1187.9 + 3(1x3726.3)^2}$	6,877.5 $lbf \cdot in$
$d_{Axle,Min}$	Minimum Axle Diameter	$(\frac{16n_{Axle,Min}}{\pi}(\frac{A/1000}{S_e} + \frac{B/1000}{S_{Ult,S}}))^{1/3} = (\frac{16(1.25)}{\pi}(\frac{4103.3/1000}{23.1765} + \frac{6877.5/1000}{34}))^{1/3}$	1.092 in

τ_{Axle}	Torsional Stress on Axle	$\frac{16(T_{Max})}{\pi d^3} = \frac{16(5962.1)}{\pi(1.25)^3}$	15.5467 kpsi
σ_{Axle}	Tensile Bending Stress on Axle	$\frac{32(M_{Max})}{\pi d^3} = \frac{32(1866.3)}{\pi(1.25)^3}$	9.7333 kpsi
V_{ST}	Vertical Suspension Travel	$(L_{LA})(cos(\theta_{Max,r}) - cos(\theta_{Min,r})) = (24)(cos(0.925) - cos(0.8203))$	1.9244 in
Fixed Coupling Equations			
d_{Bolt}	Major Diameter of Fixed Coupling Bolts	Chosen	0.25 in
$S_{y,2}$	Yield Strength for ¼ in SAE Grade 2 Steel Bolt	SAE Bolt Strength Table	57 kpsi
N_B	Number of Bolts	Chosen from $N_{B,Min}$	4
D_{BC}	Diameter of Bolt Circle	$D_{Out} + 1.25 = 1.625 + 1.25$	2.875 in
$D_{BC,0}$	Outer Diameter of Bolt Circle	$D_{Out} + 2.0 = 1.625 + 2.0$	3.625 in
r_{BC}	Radius of Bolt Circle	$\frac{D_{BC}}{2} = \frac{2.875}{2}$	1.4375 in
F_{BC}	Force on Bolt Circle	$\frac{T_{max}}{r_{BC}} = \frac{5962.1}{1.4375}$	4,147.5 lbf
A_{BCS}	Cross-Sectional Area of Bolt	$\frac{\pi d^2}{4} = \frac{\pi(0.25)^2}{4}$	0.0491 in^2
$F_{S,Bolt}$	Shear Force on Bolt	$(0.5)(S_{y,2}x1000)(A_{BCS}) = (0.5)(57x1000)(0.0491)$	1,399 lbf
$N_{B,Min}$	Lowest Number of Bolts Required	$\frac{F_{BC}}{F_{S,Bolt}} = \frac{4147.5}{1399}$	2.9647
xy	Bolt Positioning in Square Pattern	$\frac{r_{BC}}{\sqrt{2}} = \frac{1.4375}{\sqrt{2}}$	1.0165
P_t	Shear Load Per Bolt	$\frac{T_{Max}}{(N_B)(r_{BC})} = \frac{5962.1}{(4)(1.4375)}$	1,036.9 lbf

P_{ex}, P_{ey}	Eccentric Load in Square Pattern	$\frac{T_{Max}}{4(r_{BC})^2} = \frac{5962.1}{(4)(1.4375)^2}$	721.3105 lbf
P_x	Total Shear Load Component, X	$P_{ex} + P_{cx} = 721.3105 + 0$	721.3105 lbf
P_y	Total Shear Load Component, Y	$P_{ey} + P_{cy} = 721.3105 + 0$	721.3105 lbf
P_S	Total Shear Load	$\sqrt{P_x^2 + P_y^2} = \sqrt{721.3105^2 + 721.3105^2}$	1,020.1 lbf
n_{BC}	Mounting Bolts Factor of Safety	$\frac{F_{BC}}{P_S} = \frac{4147.5}{1020.1}$	4.0659 > 1.75

References:

1. Nisbett, J. Keith. Shigley's Mechanical Engineering Design. 12th ed., McGraw-Hill Education, 2024.
2. Ayala, Ricardo, STEEL ALLOYS, ME 314, Canvas
3. Ayala, Ricardo, BOLT DIMENSIONS, ME 314, Canvas
4. Ayala, Ricardo, Symmco_Oilite, ME 314, Canvas
5. Ayala, Ricardo, ME314 - ETHICS, ME 314, Canvas
6. Ayala, Ricardo, NA1149 WASHER, ME 314, Canvas
7. Ayala, Ricardo, ANSI B92-1 8-16 SPLINE PITCH, ME 314, Canvas
8. Ayala, Ricardo, ME314 SPLINE SHAFTS, ME 314, Canvas
9. Ayala, Ricardo, SAE BOLT STRENGTH, ME 314, Canvas